

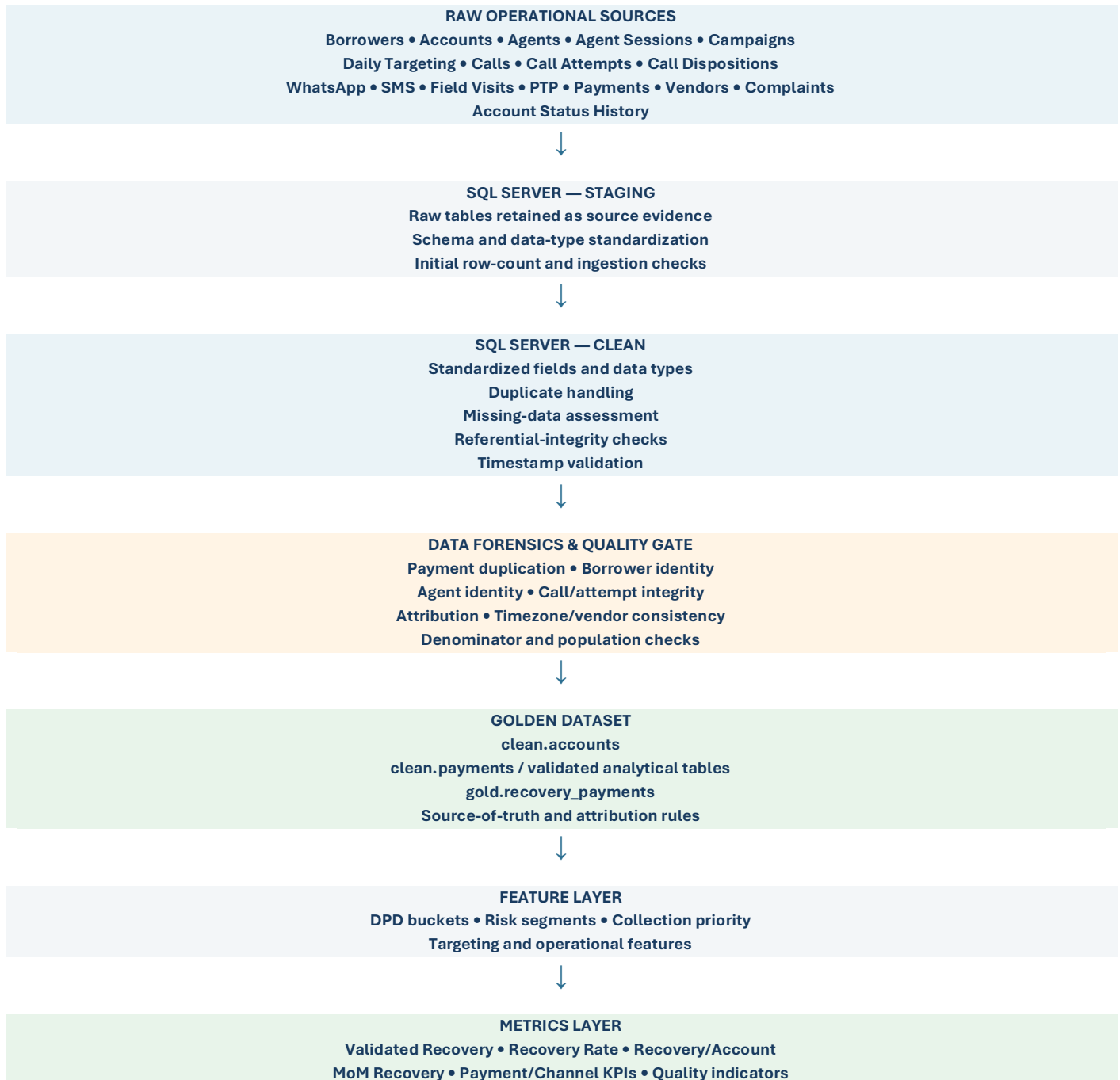
Production Analytics Architecture

Collections Recovery — Raw Data to Executive Decision

1. Architecture Objective

Design a production-ready analytical flow that converts multiple operational collections sources into governed recovery metrics and an executive Power BI dashboard. The design follows the assignment requirement: Raw → Staging → Clean → Golden → Feature → Metrics → Dashboard.

2. End-to-End Flowchart



**CONSUMPTION**

Python: independent validation and statistical analysis
Power BI: executive dashboard and drill-down analysis

**LEADERSHIP DECISION**

Performance assessment • Collection prioritization
Controlled experiments • Investment decision

3. Core Data-Quality & Governance Controls

Control	Where it operates	Purpose
Data contracts & schema checks	Ingestion / Staging	Detect changed columns, data types and unexpected source structures.
Primary-key uniqueness	Staging / Clean	Identify duplicate IDs before downstream metrics.
Referential integrity	Clean / Golden	Ensure payments/events connect to valid accounts and entities.
Duplicate detection	Clean / Golden	Prevent duplicate payment or digital-event inflation.
Timestamp & timezone validation	Clean / Forensics	Prevent incorrect day/hour classification.
Entity resolution	Forensics / Golden	Control borrower, agent and event-identifier ambiguity.
Payment reconciliation	Golden / Metrics	Ensure only validated successful payments drive recovery.
Metric versioning & lineage	Metrics / BI	Make KPI definitions traceable and reproducible.
Monitoring & anomaly detection	Metrics / Production	Flag unusual row counts, status mix or recovery movements.

4. Incremental Processing, Late Data & Backfills

- Incremental loads should process newly arrived or changed source records using source event timestamps and stable identifiers.
- Late-arriving events should be merged into the affected historical partition and trigger recalculation of impacted daily/monthly metrics.
- Backfills should be version-controlled and re-run through the same transformation logic so historical corrections remain reproducible.
- Golden and metric outputs should carry load/run timestamps and lineage identifiers to support auditability.

5. Golden Dataset Design

Primary recovery output: gold.recovery_payments

Validated recovery is generated from the cleaned payment/account layer using the approved payment attribution path. The Golden layer is then consumed consistently by Python and Power BI.

Golden principle	Implementation
Account source	clean.accounts
Payment-quality source	clean.payments
Validated recovery source	gold.recovery_payments
Payment identifier	payment_id after duplicate handling
Borrower attribution	payment.account_id → accounts.account_id → accounts.borrower_id
Call event key	Validated attempt_id preferred; call_id retained as source/reference field

6. Business Outputs Supported

Output	Purpose / observed result
Validated recovery	₹1.316B validated recovery; 17,534 validated recovery transactions
Portfolio	30,000 accounts; ₹10.489B outstanding; average DPD 56.51 days
Monthly performance	Independent month-on-month recovery calculation used to test the 11% claim
Risk / DPD	Recovery and exposure by risk segment and DPD bucket
Data-quality reporting	Duplicate, identity, attribution and reconciliation findings
Executive decision support	Power BI dashboard and controlled-investment recommendation

7. Production Design Principles

- Raw data remains immutable and auditable.
- Cleaning rules are deterministic, documented and reproducible.
- Golden data contains only validated entities/transactions used for management KPIs.
- Every KPI is traceable from Power BI to a defined metric and Golden source.
- Identity conflicts are flagged rather than silently overwritten.
- Data-quality failures block or quarantine affected downstream metrics where material.
- Operational anomalies are monitored continuously and investigated before executive reporting.

8. Architecture Conclusion

The proposed architecture separates source evidence from analytical truth and creates a governed path from operational data to executive decision-making. SQL Server acts as the core transformation and quality-control layer; the Golden Dataset becomes the common source for Python validation and Power BI reporting.

Design objective: make recovery metrics reproducible, traceable, monitored and resilient to duplicates, identity issues, late-arriving data and changing source structures.